Bohr's Atomic Model

Project Report

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MIS AND CM

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in

Artificial Intelligence and Data Science

by

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Certificate

This is to certify that the report entitled **Bohr's Atomic Model** submitted by PUNIT LOHAN (DL.AI.U4AID24127), ALAMPALLY KARTHIK (DL.AI.U4AID24104), Tamilmani Saraswathi Sathyaa (DL.AI.U4AID24135), and VEPURI SATYA KRISHNA (DL.AI.U4AID24140) to the School of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Faridabad, in partial fulfilment of the B.Tech degree in Artificial Intelligence and Data Science, is a Bonafide record of the project work carried out by them under our guidance and supervision.

This report has not been submitted to any other University or Institute for any purpose.

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Declaration

We hereby declare that the project report titled **Bohr's Atomic Model**, submitted for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology of the School of AI, Amrita Vishwa Vidyapeetham, Faridabad, India, is a Bonafide work done by us under the supervision of Dr.Rakesh , Dr. Mrritunjoy and Dr. Jaipraksh. This submission represents our ideas in our own words and where ideas or words of others have been included.
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Abstract

This report explores the design and simulation of **Bohr's Atomic Model for Hydrogen Atom**. The study focuses on the mathematical and computational principles underlying the quantized orbits of electrons around the nucleus. By leveraging MATLAB for dynamic simulation, the project visualizes electron motion in orbits, highlighting the relationships between orbital radii, velocity, and angular velocity. The simulation provides an interactive representation of hydrogen atom structure.

CONTENTS

Chapter 1:

Introduction

Atoms, the fundamental building blocks of matter, exhibit complex structures that have intrigued scientists for centuries. Among the many models proposed to explain atomic structure, Bohr's atomic model stands out as a cornerstone in understanding atomic behaviour and spectra. By incorporating principles of classical, the Bohr model provides a simplified yet insightful representation of atomic systems.

In this project, we focus on simulating Bohr's atomic model to visualize and analyse its physical and mathematical underpinnings. Emphasizing the interplay between physics and mathematics, this study explores critical aspects.

To achieve this, we leverage **MATLAB**, a versatile computational platform widely used for numerical modelling, simulation, and data visualization. MATLAB's capabilities, including its mathematical toolboxes and graphical interfaces, make it an ideal choice for simulating electron orbits.

By combining theoretical principles with MATLAB-based simulations, this project aims to enhance the understanding of Bohr's atomic model and demonstrate the effectiveness of computational tools in studying fundamental scientific concepts.

Chapter 2:

Bohr's Postulates:

2.1 Postulates of Bohr

1. The centre of the atom is called nucleus.
2. Electron revolves only in fixed orbits with fixed energy and fixed velocity.
3. Quantization condition: electron can only revolve in those circular orbits for which the angular momentum (L) is an integral multiple of h/2π.
   L=nh/2π
4. L = r x p r= (radius) vector rep of angular momentum
   L = r x mv p = mv (momentum)
   mvrsin*θ = mvrsin90° = mvr
   L= mvr
   n= shell number
   m = mass of electron
   v = velocity of electron
   r = radius of that orbit
   h = planks constant (6.636 × 10*-34 Js)
   mvr = nh/2π
5. While revolving the electrostatic force between electron and nucleus provides centripetal force.

$$F_e = k_e \frac{q_1 q_2}{r^2}$$

where

- F_e is the force
- k_e is the Coulomb's constant (8.987×10^9 $N.m^2.C^{-2}$)
- q_1 and q_2 are the signed magnitudes of the charges
- r is the distance between the charges

$$F_c = \frac{mv^2}{r}$$

6. Number of protons = atomic number

   F(electrostatic)=F(centripetal)

$$K\frac{q1.q2}{r^2}=\frac{m.v^2}{r}\ \rightarrow I\ \ (K=\frac{1}{4\pi\varepsilon})$$

7. Radius of the shell:

$$L=m.v.r=\frac{n.h}{2\pi}\ \ \rightarrow II$$

$\frac{II^2}{I}$:

$$r=(0.529).\frac{n^2}{z}$$

8. Velocity of the electron in its atomic shell:

$$m.v.r=\frac{n.h}{2\pi}$$

$(\boldsymbol{By\ substituing\ r})$

$$v=(2.18).10^6.\frac{z}{n}(ms^{-1})$$

Chapter 3

Mathematical Analysis

The underlying concepts used were:

1. Matrix Operations
Lower Triangular Matrix (tril):
The matrix T is generated using tril(ones(length(ref_size))), creating a lower triangular matrix of ones. This matrix is used to cumulatively sum angular velocities, effectively simulating the electron's rotational motion. This is our transformation matrix of size (10001x10001).

```
transformation matrix
[1, 0, 0, 0, 0,.....;
 1, 1, 0, 0, 0,.....;
 1, 1, 1, 0, 0,.....;
 .   .            .....;
 .   .            .....;
 .   .                1]
```

Matrix Multiplication:
theta = T * W computes the cumulative angle of rotation by multiplying the triangular matrix with a constant angular velocity vector, simulating continuous angular displacement over time.

Coordinate Transformation:
The transformation of polar coordinates to Cartesian coordinates is done through:
x=cos(θ), y=sin(θ)

The matrix multiplication R * xy scales the unit circle to the Bohr radius.

2. Trigonometry
Circular Motion:

The code uses trigonometric functions (cos and sin) to model the electron's circular orbit around the nucleus:
$x(t) = r \cdot \cos(\theta)$, $y(t) = r \cdot \sin(\theta)$

This represents uniform circular motion in two dimensions.
3. Classical Mechanics
Angular Velocity and Linear Velocity Relation:
The angular velocity w is derived from the linear velocity v and radius r:

w= v/r
This reflects the relationship between circular and linear motion.

Electron Path Simulation:
The electron's position is updated iteratively using for loop to simulate motion along the orbit using its angular displacement.

4. Vector and Scalar Multiplication
Scaling of Coordinates:
The scaling matrix R scales the unit circle to the actual Bohr radius, transforming the normalized circular path to the physical orbit size.

5. Numerical Simulation
Discrete Time Steps:
The loop iterates through discrete time steps (pause (0.1)) to animate the electron's continuous motion along the orbit.
These mathematical concepts collectively simulate the behaviour of an electron orbiting the nucleus in Bohr's atomic model.

Chapter 4

Simulation Analysis

4.1 MATLAB Code

Photo:

```
close all;    % Close all figure windows
clear;        % Clear workspace
clc;          % Clear command window

ref_size = 0:0.01:100;  % Define reference size range

% Create lower triangular matrix of ones
T = tril(ones(length(ref_size)));

% Define constants
r = 0.529 * 10^-10;  % Bohr radius in meters
v = 2.18 * 10^6;     % Electron velocity in m/s
w = v / r;            % Angular velocity

% Define angular velocity vector
W = w * (ones(length(ref_size), 1));

% Calculate angle vector
theta = T * W;

% Calculate x and y components of the electron's motion
x = cos(theta');
y = sin(theta');

% Combine x and y into a single matrix
xy = [x; y];

% Define the radius matrix
R = [r 0; 0 r];

% Path of the electron
Path = R * xy;

% Plot position of the proton
plot3(0, 0, 0, 'r.', 'MarkerSize', 20);
hold on;
```

```
% Plot initial position of the electron
Electron = plot3(r, 0, 0, 'k.', 'MarkerSize', 20);
hold on;

grid on;

% Set axis limits
xlim([-1.5 * r, 1.5 * r]);
ylim([-1.5 * r, 1.5 * r]);
zlim([-1.5 * r, 1.5 * r]);


% Add axis labels
xlabel('X (m)');
ylabel('Y (m)');
zlabel('Z (m)');


% Loop through reference size and animate the electron's path
for i = 1:length(ref_size)
    plot3(Path(1, i), Path(2, i), 0, 'b.', 'MarkerSize', 0.0001);  % Plot electron path
    pause(0.1);      % Pause to create animation effect
    Electron.XData = Path(1, i);
    Electron.YData = Path(2, i);
    Electron.ZData = 0;
end
```

Code:

```
close all;   % Close all figure windows

clear;       % Clear workspace

clc;         % Clear command window


ref_size = 0:0.01:100;  % Define reference size range


% Create lower triangular matrix of ones

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% Define constants
```

```
r = 0.529 * 10^-10;  % Bohr radius in meters
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x = cos(theta');
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% Combine x and y into a single matrix
xy = [x; y];

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R = [r 0; 0 r];

% Path of the electron
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% Plot position of the proton
plot3(0, 0, 0, 'r.', 'MarkerSize', 20);
hold on;

% Plot initial position of the electron
Electron = plot3(r, 0, 0, 'k.', 'MarkerSize', 20);
hold on;

grid on;

% Set axis limits
xlim([-1.5 * r, 1.5 * r]);
ylim([-1.5 * r, 1.5 * r]);
zlim([-1.5 * r, 1.5 * r]);

% Add axis labels
xlabel('X (m)');
ylabel('Y (m)');
zlabel('Z (m)');

% Loop through reference size and animate the electron's path
for i = 1:length(ref_size)
    Electron.XData = Path(1, i);
```

```
    Electron.YData = Path(2, i);

    Electron.ZData = 0;

    plot3(Path(1, i), Path(2, i), 0, 'b.', 'MarkerSize', 0.001);  % Plot electron path

    pause(0.1);  % Pause to create animation effect

end
```

4.2 MATLAB Code Explanation:

- **Transformation Matrix**: Using the tril() function in MATLAB
  - **T =** tril(ones(length(t)))
- **Radius of orbital** (r): 0.529*10^-10 m
- **Velocity of electron** (v): 2.18*10^-10 m/s
- **Angular Speed**:
  - ω: v/r
  - ω = 4.1210 rad/s
- **Theta Matrix**:
  - **θ =** ω*t
  - **theta =** T*W
- **Position Matrix (Initial):** The initial position matrix contains two vectors “x” and “y”:
  - x = cos(theta’)
  - y = sin(theta’)
  - xy = [x ; y]
- **Radius Matrix**: The radius matrix contains the value of radius in a diagonal matrix.
  - **R =** [r 0 ; 0 r]
- **Position Matrix (Final):** The final position matrix contains the multiplication of radius matrix and the initial position matrix
  - **Path =** [R*xy]
- **Plotting the Nucleus:** Using the plot3() function to plot the nucleus
  - plot3(0,0,0, ‘r.’, ‘MarkerSize’, 20)
- **Plotting the Electron:** Using the plot3() function to plot the initial position of the electron.
  - Electron = plot3(r,0,0, ‘k.’, ‘MarkerSize’, 20)
- **Limiting the Plot:** Using the lim() function to show a limited view of the path for better visibility.

    - xlim ([-1.5*r,1.5*r])
    - ylim([-1.5*r,1.5*r])
    - zlim([-1.5*r,1.5*r])
- **Animation:** Using for loop in MATLAB to plot the animation of the electron moving in its orbit.
    - for i=1:length(ref_size)
    -     Electron.XData=Path(i,1);
    -     Electron.YData=Path(i,2);
    -     Electron.ZData=0;
    -     plot3(Path(i,1),Path(i,2),0,'b.','MarkerSize',0.001)
    -     pause(0.1)

where,

- Electron.XData=Path(i,1); updates the x-position of the electron
- Electron.YData=Path(i,2); updates the y-position of the electron
- plot3(Path(i,1), Path(i,2) ,0, 'b.', 'MarkerSize', 0.001); plots the path of the orbit.

Chapter 5

Results and Discussion

7.1 Simulation Results

The MATLAB simulation successfully demonstrates the circular orbit of an electron around the nucleus in Bohr's atomic model. The transformation matrices, mathematical calculations, and animation represent the quantized motion of the electron.

7.2 Significance of Results

- Electron Trajectory: The simulation replicates the uniform circular motion of an electron as described in Bohr's model.
- Angular Velocity: Derived from the relationship between linear velocity and radius, angular velocity validates theoretical predictions.
- Visualization: The path traced by the electron aligns with expected quantized orbits.

7.3 Limitations:

-Bohr model: Bohr model is valid only for H like species i.e. single electron species.

- Bohr's model is not applicable to multi-electron atoms because it neglects electron-electron repulsion. It assumes circular orbits and simple quantization, which fail to describe the complexities of multi-electron systems.

Chapter 6

Conclusion

Bohr’s atomic model provides a foundational understanding of quantum mechanics, emphasizing the quantized nature of electron orbits. Through this project, we simulated the hydrogen atom using MATLAB, offering a dynamic visualization of theoretical principles. By leveraging transformation matrices and trigonometric functions, the simulation enhances comprehension of atomic behavior. This project demonstrates the synergy between computational tools and scientific inquiry, reinforcing MATLAB’s significance in modern research.